AGC 114905 and the Low-Response Galaxy Regime

A Resolved-Source Test of Curvature Polarization Transport Gravity

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Abstract

AGC 114905 is a gas-rich ultra-diffuse galaxy whose unusually slow rotation makes it a stringent test of how gravity responds when the measured baryonic source is substantial but the permitted dynamical enhancement is modest. In the standard dark-matter picture, the additional response is supplied by a non-baryonic halo. Modified Newtonian Dynamics (MOND) instead applies a universal low-acceleration scale. Curvature Polarization Transport Gravity (CPTG) attributes the additional response to geometric curvature polarization and transport, with a structural acceleration scale $a_\star$ determined for the gravitating system.

Using the established AGC 114905 geometry, five published tilted rings, and continuous stellar and gas surface-density laws, the registered CPTG galaxy reduction yields

$$a_\star = 7.2797 \times 10^{-12}\ \mathrm{m\,s^{-2}}, \qquad \chi^2_{\mathrm{CPTG}} = 4.046,$$

corresponding to $\chi^2_\nu = 1.011$ for four degrees of freedom. Every CPTG residual is smaller than 1.5σ. On the same reconstructed galaxy, baryon-only dynamics gives $\chi^2 = 50.504$, while MOND with the simple interpolation function $\mu(x) = x/(1+x)$ and universal $a_0 = 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ gives $\chi^2 = 672.329$.

A 10,000-draw coupled uncertainty analysis preserves the CPTG diagnostic preference over both comparators in 99.98% of realizations. The retained field also gives the post-solution Structural Mode $N = 1.4967$, identifying AGC 114905 as a low-mode CPTG structure. The result is a strong low-response validation of the CPTG galaxy law: the observed baryonic source together with the theory's geometric response reproduces the resolved rotation curve to near-ideal error-weighted precision, while the galaxy's weak response is encoded by its own structural acceleration scale rather than by a universal acceleration constant.

1 AGC 114905 as a Gravity Test

AGC 114905 is a gas-rich ultra-diffuse galaxy: it contains a large, spatially extended baryonic disc but has a very low stellar surface brightness and an unusually slow resolved rotation curve. Its H I velocity field is sufficiently regular to permit a tilted-ring reconstruction, while the optical and gas data provide an independently measured baryonic source [1, 2]. Those two properties make the galaxy unusually informative. The matter distribution can be measured directly, and the observed circular speed leaves comparatively little room for an additional gravitational contribution.

This matters beyond the dynamics of one galaxy because galaxy rotation curves sit at the intersection of cosmological structure formation and theories of gravity. In the standard dark-matter picture, galaxies form inside non-baryonic haloes whose gravitational field supplements that of the observed stars and gas. A galaxy with substantial baryonic mass but unusually low circular speed therefore requires an unusually weak halo contribution over the measured region. The 2024 analysis of AGC 114905 found that conventional cold-dark-matter fits occupy an unusually low-density part of halo parameter space, while alternative dark-sector constructions can remain viable [2]. The system consequently probes the small-scale halo structure expected from the cosmological dark-matter framework.

MOND asks a different question. Rather than adding a dark halo, it modifies the low-acceleration relation between the baryonic field and the observed dynamics. In its standard galaxy application, the transition is controlled by a universal acceleration constant a_0 of order $10^{-10}\ \mathrm{m\,s^{-2}}$ [4, 5]. AGC 114905 is especially demanding for such a universal enhancement because the galaxy lies deeply in the low-acceleration regime but its measured rotation curve does not permit a correspondingly large boost. At the established source geometry, the simple-interpolation MOND prediction therefore lies far above the measured circular speeds [2].

CPTG takes a third route. Ordinary baryonic matter is the material source of the galaxy field, while the additional gravitational response is carried by curvature polarization and curvature transport [6]. The galaxy reduction has a fixed functional form and locked source weights across systems. Galaxy-to-galaxy response strength is represented by the structural acceleration scale $a_\star$, an object-dependent quantity inferred from the resolved source and kinematics. The associated curvature-transport radius R_c then follows from the CPTG structural relation.

After the dynamical solution is retained, CPTG evaluates a second quantity: Structural Mode N. This continuous post-solution coordinate describes how the polarization–transport response is organized across radius. In simple terms, $a_\star$ characterizes response strength, while N characterizes radial organization.

The *Curvature Structure Mode Index* (CSMI) is the catalog layer built from N [7]. It translates intervals of the continuous structural coordinate into descriptive classes derived from the solved CPTG field. The classification therefore follows the dynamics rather than an externally supplied Hubble type or visual morphology.

The paper proceeds from source to solution. Section 2 establishes the common observational reconstruction. Section 3 summarizes the CPTG galaxy response and the role of $a_\star$. Section 4 presents the rotation-curve result and the baryon/MOND comparison. Section 5 propagates the observational uncertainties, and Section 6 interprets the retained field through N and CSMI.

2 The Common Observational Source

Baryon-only dynamics, simple-interpolation MOND, and CPTG are evaluated on one common reconstruction of AGC 114905. The adopted distance, inclination, radial coordinates, velocity uncertainties, and baryonic source therefore define a shared observational basis for the three calculations.

2.1 Geometry and five-ring kinematics

The adopted AGC 114905 geometry is

$$D = 78.7 \pm 1.5 \text{ Mpc}, \qquad i = 31 \pm 2^\circ, \tag{1}$$

and the resolved circular-speed field is represented by five independent tilted rings [2]. These five rings constitute the kinematic data set used throughout the paper; the continuous radial reconstruction is reserved for the baryonic source.

Because AGC 114905 is close to face-on, the inclination matters directly:

$$V_{\rm circ} \propto \frac{1}{\sin i}. \tag{2}$$

This adopted geometry is carried unchanged into the baryon-only, simple-interpolation MOND, and CPTG calculations.

2.2 Continuous stellar and gas source

The kinematics contain five independent rings, but the baryonic source is known as a continuous radial distribution. The 2024 analysis provides fitted stellar and gas surface-density laws and states that these functions are used in the mass modelling [2]. In angular radius θ (arcsec), the stellar surface density is

$$\Sigma_\star(\theta) = 3.6703\, e^{-\theta/3.6155} \left(1 + 0.55776\theta - 0.1126\theta^2 + 0.0085\theta^3 - 0.0002\theta^4\right) \ \mathrm{M_\odot\, pc^{-2}}, \tag{3}$$

and the gas mass-model profile is

$$\Sigma_{\rm gas}(\theta) = 3.953\, e^{-\theta/0.676} \left(1 + \frac{\theta}{375.220}\right)^{559.231} \ \mathrm{M_\odot\, pc^{-2}}. \tag{4}$$

The gas source is the H I+helium mass-model component used by the observational analysis, which already incorporates the documented helium correction [1, 2].

The component force curves use the same vertical-law families as the source analysis: a sech^2 stellar distribution and a Gaussian gas distribution. At the fiducial geometry the corresponding scale heights are $z_\star \simeq 0.271$ kpc and $z_{\rm gas} = 1.3$ kpc [2].

2.3 Independent source-force verification

The baryonic force reconstruction is numerically cross-checked in two independent ways. One calculation uses a finite-thickness Hankel transform. The other directly implements the public GALPYNAMICS Cuddeford-style midplane circular-speed integral with the same normalized vertical laws. Across the five measured rings, the maximum difference between the two independent realizations is

$$0.0739\% \quad \text{for the stellar component,} \qquad 0.3319\% \quad \text{for the gas component.} \tag{5}$$

The GALPYNAMICS-equivalent realization is used for the ring-level values reported below.

The resulting common source table is shown in Table 1. The ordinary baryonic velocity is defined conventionally by

$$V_{\rm bar}^2 = V_\star^2 + V_{\rm gas}^2. \tag{6}$$

R (kpc)	$V_{\rm obs}$	σ_V	$V_\star$	$V_{\rm gas}$	$V_{\rm bar}$
1.1044	11.3016	2.1734	6.3152	3.0206	7.0004
3.3107	19.1348	2.1734	9.1411	9.1128	12.9075
5.5158	25.8542	2.1281	8.2223	14.3609	16.5481
7.7221	28.0639	2.5809	7.3914	18.1159	19.5657
9.9272	27.5839	2.2186	6.3374	20.1770	21.1489

Table 1: Common AGC 114905 source and kinematic table. All velocity quantities are in $\mathrm{km\,s^{-1}}$. $V_\star$ and $V_{\rm gas}$ are reconstructed from the published continuous surface-density laws. The same radii, observed circular speeds, uncertainties, and baryonic source are supplied to every gravity comparison in this paper.

With the observational source established, the next section summarizes how CPTG maps that source into a predicted rotation curve.

3 CPTG Galaxy Response

In the CPTG galaxy reduction, the measured baryonic structure supplies the material source and the excess over the direct baryonic field is carried by geometric curvature polarization and transport [6]. The complete galaxy-limit derivation, source mapping, transport relation, nonlinear response law, and numerical solution procedure are given in the core theory and dedicated galaxy-calculation paper [8]. The present application requires only the main structural sequence.

For a bulgeless galaxy, the locked CPTG source map combines the gas and stellar-disc contributions with coefficients 27/50 and 23/50, respectively. These coefficients are fixed by the CPTG galaxy reduction and do not vary from galaxy to galaxy [8]. The resulting source is then evaluated together with the galaxy's object-dependent structural acceleration scale $a_\star$.

The quantity $a_\star$ characterizes the strength of the CPTG response of the gravitating system. Each galaxy therefore carries its own structural acceleration scale, determined from its resolved source and kinematics. This object-dependent role distinguishes $a_\star$ from MOND's universal a_0.

For each trial $a_\star$, the CPTG structural relation determines R_c. The baryonic source, $a_\star$, and R_c then define the implicit galaxy field, which is solved at the measured radii and converted to the predicted circular-speed curve [6, 8].

The inference procedure is therefore

$$\begin{gathered}\text{observed gas and stellar source}\\ \downarrow\\ \text{fixed CPTG source mapping}\\ \downarrow\\ \text{trial object-dependent } a_\star\\ \downarrow\\ \text{derived } R_c\\ \downarrow\\ \text{fixed nonlinear CPTG galaxy solve}\\ \downarrow\\ V_{\mathrm{CPTG}}(r),\end{gathered}\tag{7}$$

with the retained $a_\star$ selected by the error-weighted agreement between the predicted and observed rotation curve.

Across galaxy applications, the measured source and inferred $a_\star$ identify the individual system, while the CPTG galaxy reduction itself remains fixed. AGC 114905 therefore tests the same response architecture used for other resolved rotation curves.

4 Rotation-Curve Result

The inferred structural acceleration scale is

$$a_\star = 7.2796939 \times 10^{-12}\ \mathrm{m\,s^{-2}}. \tag{8}$$

For this value,

$$\chi^2_{\mathrm{CPTG}} = 4.04598. \tag{9}$$

Because five measured rings constrain one inferred CPTG quantity, the fit has four degrees of freedom and therefore

$$\chi^2_{\nu,\mathrm{CPTG}} = \frac{4.04598}{4} = 1.01150. \tag{10}$$

This is a near-ideal error-weighted fit. The RMS velocity residual is 1.964 $\mathrm{km\,s^{-1}}$, and every ring lies within 1.453σ of the CPTG prediction.

The fitted $a_\star$ implies $R_c \simeq 6.03 \times 10^4$ Gpc, making the explicit finite-R_c transport remnant dynamically negligible across the five measured rings. The low-response solution is therefore governed primarily by the mapped baryonic field, the curvature-polarization contribution, and the registered nonlinear response.

4.1 Direct comparison on the same galaxy

The ordinary baryonic source alone gives

$$\chi^2_{\mathrm{baryon}} = 50.50438. \tag{11}$$

The baryons are therefore insufficient to reproduce the observed five-ring curve at the adopted geometry.

For the MOND comparator, the same ordinary baryonic source is evaluated with the simple interpolation function

$$\mu(x) = \frac{x}{1+x}, \qquad a_0 = 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}. \tag{12}$$

The corresponding acceleration relation is

$$g_{\mathrm{MOND}} = \frac{g_{\mathrm{bar}}}{2}\left(1 + \sqrt{1 + \frac{4a_0}{g_{\mathrm{bar}}}}\right), \tag{13}$$

which gives

$$\chi^2_{\mathrm{MOND}} = 672.32940. \tag{14}$$

The resulting MOND enhancement is much too large for this galaxy at the established source geometry.

R (kpc)	$V_{\rm obs}$	$V_{\rm bar}$	$V_{\rm CPTG}$	$\Delta V_{\rm CPTG}$	Δ/σ	$V_{\rm MOND}$
1.1044	11.3016	7.0004	9.8825	-1.4191	-0.6529	21.7447
3.3107	19.1348	12.9075	18.0415	-1.0933	-0.5031	38.9208
5.5158	25.8542	16.5481	23.4726	-2.3816	-1.1192	50.0580
7.7221	28.0639	19.5657	27.8937	-0.1701	-0.0659	59.2066
9.9272	27.5839	21.1489	30.8064	+3.2225	+1.4525	65.4566

Table 2: Five-ring comparison at the common AGC 114905 source geometry. All velocity quantities, including $\Delta V_{\rm CPTG}$, are in $\mathrm{km\,s^{-1}}$; Δ/σ is dimensionless. CPTG uses the galaxy reduction summarized in Section 3; the MOND column uses the simple-interpolation prescription defined above.

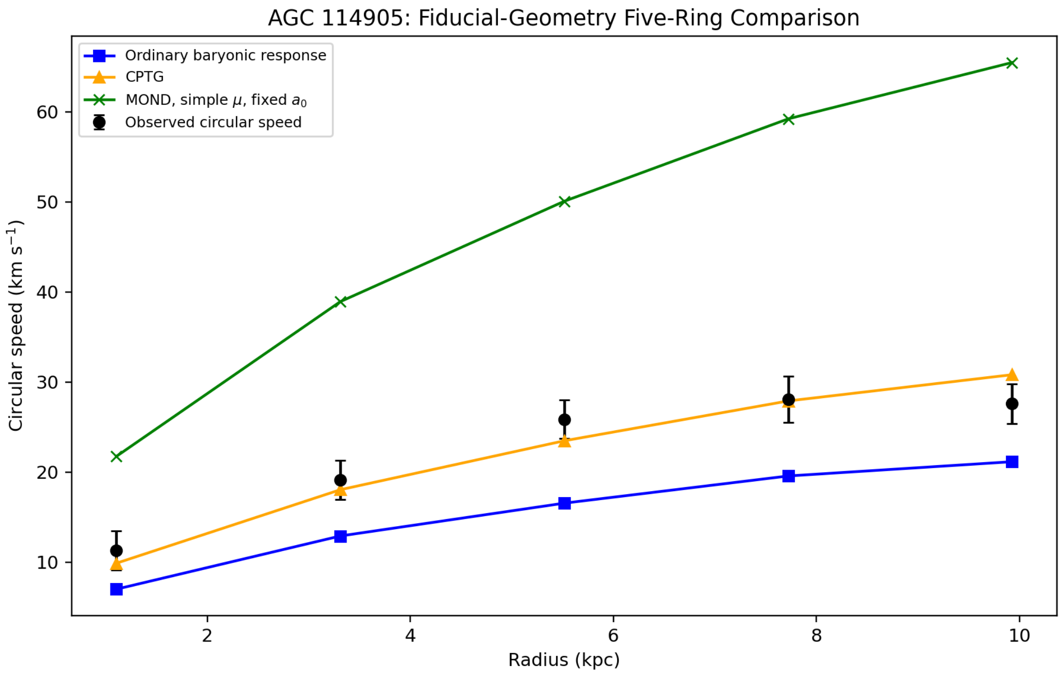

Figure 1: AGC 114905 at the adopted source geometry. Observed circular speeds are shown with error bars. The ordinary baryonic response is the solid blue curve, CPTG is the solid orange curve, and simple-interpolation MOND is the solid green curve. The CPTG curve follows all five measured rings with $\chi^2_\nu = 1.011$; baryon-only remains too low, while simple-interpolation MOND produces a much larger low-acceleration enhancement than the data permit.

Figure 1 is the central empirical result. The CPTG curve follows the five measured rings with $\chi^2_\nu = 1.0115$, an error-weighted agreement essentially at the ideal unit-reduced-χ^2 level. The largest residual is only 1.453σ. This agreement is produced by the registered CPTG galaxy response acting on the measured baryonic source, with the system's low response encoded by $a_\star$.

The next question is how this low-response solution propagates through the observational uncertainty budget.

5 Robustness under Observational Uncertainty

The uncertainty analysis propagates the measured inputs through the same CPTG response calculation. Distance and inclination are drawn from

$$D \sim \mathcal{N}(78.7, 1.5)\ \mathrm{Mpc}, \qquad 74.2 \le D \le 83.2, \tag{15}$$

$$i \sim \mathcal{N}(31, 2)^\circ, \qquad 25^\circ \le i \le 37^\circ, \tag{16}$$

The source and velocity draws are specified explicitly. A coherent stellar normalization factor is generated as

$$f_\star = 10^{\delta_\star}, \qquad \delta_\star \sim \mathcal{N}(0, 0.15), \tag{17}$$

with the scatter in dex. The coherent gas normalization is drawn as

$$f_{\mathrm{gas}} \sim \mathcal{N}\left(1, \frac{0.11}{1.04}\right), \tag{18}$$

with the numerical positivity safeguard $f_{\mathrm{gas}} \ge 0.05$. For each ring j, the observed circular speed is perturbed by an independent $z_j \sim \mathcal{N}(0, 1)$ draw after the common inclination deprojection,

$$V_{\mathrm{obs},j} = \bar{V}_{\mathrm{obs},j}(i) + z_j\, \sigma_{V,j}(i). \tag{19}$$

Shared quantities are varied coherently, with physical radii, source normalizations, deprojected velocities, and source curves recomputed for each sampled realization. The covariance matrix of the fitted stellar and gas profile coefficients is not published and is therefore not sampled; independent coefficient perturbations are not introduced.

For every realization the continuous physical source is rebuilt, $a_\star$ is re-inferred, R_c is re-derived, and the nonlinear CPTG field is solved again. Separate 5000-draw branches isolate geometry, source normalization, and measurement effects. A combined branch contains 10,000 realizations.

The three models are compared draw by draw on the same sampled galaxy reconstruction. Baryon-only carries no fitted gravity parameter; simple-interpolation MOND retains its universal fixed a_0; CPTG re-infers the object-dependent $a_\star$ specified by its galaxy reduction. To account for that one inferred CPTG quantity, the matched-source five-ring diagnostic is

$$\mathrm{BIC}_{\mathrm{diag}} = \chi^2 + k \ln 5, \tag{20}$$

with $k = 1$ for CPTG and $k = 0$ for baryon-only and simple-interpolation MOND. This is a deliberately limited same-source diagnostic rather than a full Bayesian evidence calculation or a replacement for the observational posterior analysis.

Audit branch	CPTG lowest $\mathrm{BIC}_{\mathrm{diag}}$	Baryon lowest $\mathrm{BIC}_{\mathrm{diag}}$	MOND lowest $\mathrm{BIC}_{\mathrm{diag}}$	$a_\star$ p16	$a_\star$ p50	$a_\star$ p84
Geometry only	5000 (100.00%)	0 (0.00%)	0 (0.00%)	5.667×10^{-12}	7.339×10^{-12}	9.575×10^{-12}
Source normalization only	5000 (100.00%)	0 (0.00%)	0 (0.00%)	6.097×10^{-12}	7.177×10^{-12}	8.487×10^{-12}
Measurement only	5000 (100.00%)	0 (0.00%)	0 (0.00%)	5.826×10^{-12}	7.315×10^{-12}	8.997×10^{-12}
Combined	9998 (99.98%)	2 (0.02%)	0 (0.00%)	4.977×10^{-12}	7.204×10^{-12}	1.050×10^{-11}

Table 3: Coupled uncertainty audit. Each model column reports the number and fraction of matched-source realizations in which that model has the lowest five-ring diagnostic BIC. The MOND column refers specifically to the simple interpolation function $\mu(x) = x/(1+x)$ with fixed universal $a_0 = 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}$. The three $a_\star$ columns are in $\mathrm{m\,s^{-2}}$ and apply to the CPTG solutions. In the combined branch, baryon-only has the lowest diagnostic BIC in two marginal cases; simple-interpolation MOND has the lowest diagnostic BIC in none.

The combined 2.5–97.5 percentile interval is

$$3.480 \times 10^{-12} \le a_\star \le 1.510 \times 10^{-11}\ \mathrm{m\,s^{-2}}, \tag{21}$$

with median

$$a_{\star,\mathrm{med}} = 7.204 \times 10^{-12}\ \mathrm{m\,s^{-2}}. \tag{22}$$

The median diagnostic separations are

$$\Delta\mathrm{BIC}_{\mathrm{CPTG-baryon}} = -44.20, \tag{23}$$

$$\Delta\mathrm{BIC}_{\mathrm{CPTG-MOND}} = -670.77. \tag{24}$$

Across the combined 10,000 realizations, CPTG has the lowest diagnostic BIC in 9998 cases (99.98%), baryon-only in 2 cases (0.02%), and simple-interpolation MOND in none. The two baryon-only lowest-BIC cases are marginal after the one-parameter CPTG penalty, while the MOND diagnostic remains far above both alternatives. The inferred $a_\star$ distribution remains well inside the numerical search interval in every realization.

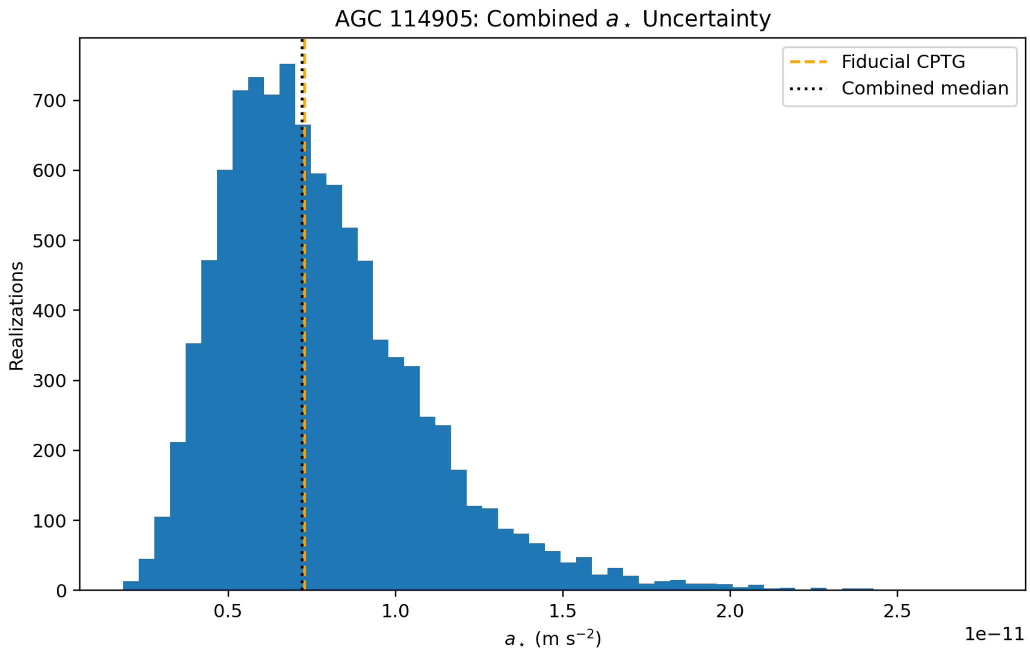

Figure 2: Distribution of the inferred CPTG structural acceleration scale over the combined 10,000-draw uncertainty audit. The fiducial value is marked in orange. The sampled distance, inclination, source normalization, and velocity uncertainties preserve the low-$a_\star$ character of the solution.

The rotation-curve result is therefore both precise at the fiducial source and stable across the declared uncertainty model. Having established the dynamical solution, the paper can now ask a different question: what kind of radial curvature organization does that solved field represent?

6 Structural Mode N and the CSMI Structural Catalog

The rotation curve establishes the strength of the CPTG response required by AGC 114905. Structural Mode N then characterizes the radial organization of that retained solution. It is a dimensionless coordinate evaluated downstream from $a_\star$, R_c, and the solved radial field [7], so the dynamical fit and the structural classification remain separate stages of the calculation.

The detailed Structural Mode functional is derived and validated in the dedicated CPTG Structural Mode study [7]. For the present paper, the important interpretation is that lower N corresponds to structural variation weighted farther outward across the registered radial domain, while progressively larger N indicates that the dominant structural variation is concentrated farther inward. Thus $a_\star$ and N encode different information: $a_\star$ characterizes the strength of the galaxy's CPTG response, whereas N characterizes the radial organization of the solved response.

For AGC 114905, the structural readout is evaluated to the outermost measured tilted-ring radius, $R = 9.92716$ kpc. The fiducial solution gives

$$N = 1.49666.$$

The numerical readout is stable: changing the resolved radial sampling from 100 to 700 points changes N by less than 9.26×10^{-4}.

The same structural calculation is performed after every realization in the 10,000-draw uncertainty audit. The resulting distribution has $N_{2.5} = 1.48099$, $N_{16} = 1.50150$, $N_{50} = 1.68216$, $N_{84} = 1.92594$, and $N_{97.5} = 2.07776$. Geometry has essentially no rank correlation with N, and varying $a_\star$ alone across its 2.5–97.5 percentile interval changes the fiducial N by only 1.24×10^{-4}. The width of the N distribution is driven primarily by uncertainty in the relative stellar and gas source normalization.

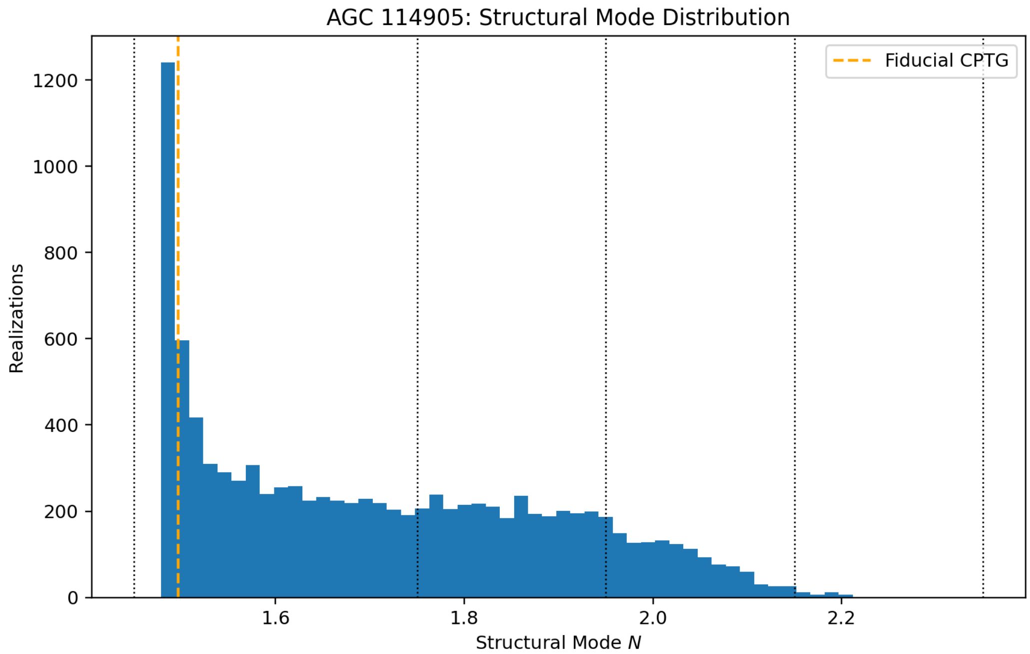

Figure 3: Continuous Structural Mode distribution for the combined uncertainty realizations. The fiducial $N = 1.4967$ solution is marked in orange. Its spread primarily traces uncertainty in the radial balance of the stellar and gas source structure.

The *Curvature Structure Mode Index* (CSMI) is the descriptive catalog constructed from the continuous N coordinate [7]. It assigns named structural bands downstream from the solved field and provides a catalog translation of the CPTG structural coordinate independent of externally supplied morphology labels.

The fiducial $N = 1.49666$ value falls in the current CSMI *Magellanic Irregular* band. Across the uncertainty realizations, 59.63% fall in that band, 27.44% in the LSB Dwarf Disk band, 12.50% in the Transition Dwarf band, and 0.43% in the LSB Spiral band. Because the continuous N value is the underlying measured structural coordinate and the categorical boundaries are more sensitive to source-composition uncertainty, this paper treats N as the primary structural result and CSMI as its secondary descriptive catalog translation.

7 Physical Interpretation

AGC 114905 is simultaneously a low-acceleration galaxy and a low-response galaxy. The distinction is physically important. Low acceleration identifies the regime; the measured rotation curve determines how much additional gravitational response that regime actually permits. For AGC 114905, that permitted enhancement is modest.

The ordinary baryonic curve lies below the observations, while simple-interpolation MOND with universal a_0 lies far above them. CPTG predicts the intermediate response required by the data. Its inferred

$$a_\star = 7.2797 \times 10^{-12}\ \mathrm{m\,s^{-2}}$$

is small, and the resulting CPTG curve tracks all five rings with

$$\chi^2_\nu = 1.0115.$$

Every residual remains within 1.5σ. These numbers make AGC 114905 a particularly strong validation of the CPTG galaxy reduction in the low-response regime.

The physical content of that fit is also clear. The material source is the observed stellar and gas distribution. The difference between the direct baryonic curve and the successful CPTG curve is carried by the theory's geometric polarization–transport response. Within CPTG, the rotation curve is therefore accounted for by the measured baryons and geometric response alone; a non-baryonic halo is unnecessary for this solution.

That statement is specific to the CPTG description and should be distinguished from the viability of dark-halo models in their own parameter spaces. The observational literature reports low-density halo realizations that can accommodate AGC 114905 [2, 3]. The significance of the CPTG result is that the observed rotation curve already has an excellent solution before introducing such a halo component.

The simple-interpolation MOND comparison is more direct because its universal acceleration scale gives a definite prediction for the same reconstructed baryonic source. At the adopted geometry, the predicted enhancement is much larger than the data permit. The inferred CPTG scale satisfies

$$\frac{a_\star}{a_0} \simeq 0.0607, \tag{25}$$

so AGC 114905's structural response scale is only about six percent of the conventional MOND acceleration scale. CPTG accommodates that weak response through the object-dependent $a_\star$ while retaining the fixed galaxy response law.

Structural Mode adds information of a different kind. The small $a_\star$ identifies a weak response amplitude, while the fiducial $N = 1.4967$ identifies a low-mode radial organization. Their separation is empirically useful: uncertainty in $a_\star$ has little direct effect on N, whereas the N distribution responds more strongly to the relative stellar/gas source structure. AGC 114905 is therefore both dynamically low-response and structurally low-mode within CPTG.

8 Conclusion

AGC 114905 is an unusually discriminating galaxy because its resolved baryonic source is substantial while its measured circular-speed field permits only a modest gravitational enhancement. Using the established $D = 78.7 \pm 1.5$ Mpc and $i = 31 \pm 2°$ geometry, the five independent tilted rings, and the published continuous stellar and gas source profiles, the CPTG galaxy reduction gives

$$a_\star = 7.2797 \times 10^{-12}\ \mathrm{m\,s^{-2}}, \qquad \chi^2_{\mathrm{CPTG}} = 4.046, \qquad \chi^2_{\nu,\mathrm{CPTG}} = 1.011.$$

All five measured rings lie within 1.5σ of the CPTG curve. On the same reconstructed galaxy, baryon-only dynamics gives $\chi^2 = 50.504$, while MOND with the simple interpolation function $\mu(x) = x/(1+x)$ and universal $a_0 = 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ gives $\chi^2 = 672.329$.

The result uses the registered CPTG galaxy response: the locked gas and stellar source weights, the established nonlinear response law, the CPTG transport relation, and one object-dependent structural acceleration scale $a_\star$. The resulting reduced chi-square is essentially unity. This is strong direct evidence that the CPTG galaxy law captures the unusually weak gravitational enhancement required by AGC 114905.

The 10,000-draw uncertainty analysis preserves the CPTG diagnostic preference over baryon-only and simple-interpolation MOND in 99.98% of realizations. The retained field gives fiducial Structural Mode

$$N = 1.4967,$$

placing AGC 114905 in the low-mode regime and supplying a structural characterization distinct from the fitted response amplitude.

The central result is therefore both dynamical and structural. The observed stellar and gas source, propagated through the CPTG geometric response, reproduces the resolved rotation curve to near-ideal error-weighted precision. Within CPTG, that agreement is achieved with baryonic matter as the material source and with the galaxy's weak response encoded by its own structural acceleration scale. AGC 114905 consequently provides a stringent and successful low-response test of the CPTG galaxy framework.

Data, Code, and Evidence Availability

The primary observational authority for AGC 114905 is Mancera Piña et al. [2], with the resolved gas-mass modelling and source construction also documented in Mancera Piña et al. [1]. The adopted distance, inclination, five-ring kinematics, stellar surface-density profile, gas surface-density profile, and vertical source prescriptions used in this manuscript follow those published observational inputs.

The complete numerical evidence package used for the manuscript is:

`CPTG_AGC114905_PublicationEvidence_20260902_r03.zip`

`SHA-256:  fc45480a7dc7a5595b24e62cb4f09045d0b55d2b00ba54196127cb4664d339e6`

The evidence package documents the resolved stellar and gas source-force reconstruction, independent source-force verification, the fiducial five-ring baryonic and CPTG solutions, the simple-interpolation MOND comparison with fixed universal a_0, the coupled distance–inclination–source–measurement uncertainty propagation, the matched-source diagnostic model comparison, the current-authority stellar outer-tail sensitivity, and the post-solution Structural Mode and CSMI analysis. The archive includes publication-facing numerical tables, executable audit scripts, numerical reports, the deterministic assembly script for the manuscript table and figures, a `README.md` describing every file, and an internal `SHA256SUMS.txt` integrity record.

The source reconstruction, CPTG field solution, uncertainty propagation, and Structural Mode calculation are deterministic numerical operations for a specified realization. Numerical calculations retain additional guard digits; manuscript values are rounded for interpretation. Supporting CPTG theory, numerical resources, and the publication evidence archive are maintained at `https://github.com/CLG2025/CPTG`.

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